



Fact sheet on applications for GCS/LRZ computing time projects on SuperMUC-NG

System Overview	SuperMUC-NG Phase 1	SuperMUC-NG Phase 2
Avail. Compute budget in this call	600 million core hours	2.1 million GPU hours
Limit for GCS Large-Scale Call	≥ 45 M Core-h	$\geq 140,000$ GPU-h
Processor	Intel Skylake, 48 cores/node	Intel Xeon Platinum 8480+ 112 cores/node
Accelerator	None	Four Intel Data Centre GPU Max 1550 accelerators per node
Nodes/Memory	6,336 Thin nodes (96GB/node) 144 Fat nodes (768GB/node)	234 compute nodes with 512 GByte memory (CPU) 128 GByte HBM2e per GPU
Total cores	311,040 CPU cores	26,208 CPU cores 936 GPUs
Interconnect	Intel OmniPath 100 GB/s	NVIDIA/Mellanox HDR Infiniband 200 GBit/s

1. Prospective users can apply for computing time on SuperMUC-NG at the GCS-JARDS web site <https://jards.gauss-centre.eu/qcshome/application>

Applications are accepted online only; applications via email will be rejected.

Depending on the amount of compute time requested, there are two types of applications:

- a) Applications for **GCS Large-Scale** Projects requiring 45 million core hours or more on SuperMUC-NG Phase 1, or 140,000 GPU-hours or more on SuperMUC-NG Phase 2 in a one-year time frame can apply twice a year at defined periods of time. The dates for the next call can be found on the [GCS webpage](#). The call deadlines are strict, thus, applications submitted after the call deadline will be rejected.
- b) Applications for **Regular** (<45 million core hours on SuperMUC-NG Phase 1 or <140,000 GPU hours on SuperMUC-NG Phase 2) and **Test** Projects (300,000 core hours on SuperMUC-NG Phase 1, or 900 GPU-h on SuperMUC-NG Phase 2) can be submitted continuously (rolling call). Please tag the appropriate box on the GCS-JARDS web site.

2. Projects with a **huge computing time** demand are explicitly encouraged to apply for a GCS Large-Scale project. If the application is not granted as a GCS Large-Scale project, it is automatically considered as a regular SuperMUC-NG project. Therefore, there are no disadvantages for the applicants submitting a large-scale proposal. Large-Scale projects receive

mentoring support by LRZ staff. Dedicated resources can be assigned to Large Scale projects, if required and justified.

3. You can simultaneously request core-hours on Phase 1 for CPU-only jobs and GPU hours on Phase 2 for GPU jobs. Please note that accounting always considers full nodes, even if you use less cores or GPUs than available on a node.
4. For applications for Phase 2 it is mandatory that you demonstrate performance and scalability of your code on the Intel Data Center Max 1550 GPUs (Ponte Vecchio). Please request a test project with a limit of 900 GPU hours to compile your code and produce the data.
5. Scientists employed at universities or research facilities in **Germany** are **eligible for application**; the nationality of the applicant ("Principal investigator, PI") does not play a role. The PI of a project must have a proven scientific record (preferable a PhD or comparable degree) and must be able to successfully accomplish the proposed tasks.

6. **Budget Policy for Project Extensions on SuperMUC-NG**

Unused compute time budgets for GCS Large-Scale Projects will be **cut off** at the start of the new granting period. This applies equally, whether the project was a regular or a large-scale project before. Alternatively, you may request a cost neutral prolongation of your project to consume the remaining budget, but no new compute time will be awarded.

If you have any questions, consult your [mentor](#) or issue a ticket at servicedesk.lrz.de.

7. For the preparation of the requested project description, please use the **template**, which is available in [Word](#), [LaTeX](#) and [PDF](#), and adhere to the [guidelines](#) given on the GCS webpage. Since the templates may change from call to call, make sure to use the latest version using the links above. Please be aware that the size of the description is limited to **18 pages** (font size 11pt) and **60 MB**.

In case you apply for a project extension, please also upload the status report (max. 10 pages) as a separate file (PDF). Please use the **template**, which is available in [Word](#), [LaTeX](#) and [PDF](#).

8. GCS Large Scale projects are awarded for 12 months, and projects need to justify in their proposal that the longest running simulation does **not exceed a total of 180 wall days** of computing time for their longest simulation campaign. Please specify the necessary wall days in your proposal.
9. Storage quotas on WORK need to be justified in the proposal if they largely exceed 10 TB per 10 M core-h compute time, or 500 TB in total.
10. Further details are given on the [SuperMUC-NG web site](#).
11. We offer online consulting on the application process during our HPC Lounge <https://doku.lrz.de/hpc-lounge-813636379.html>